Soham Sahasrabuddhe

sohams@iitb.ac.in • soham10.github.io • soham10

Education

Indian Institute of Technology, Bombay **2023 – 2027**
B.Tech in Engineering Physics with a minor in Mathematics *CPI: 9.72/10*

Research Experience

Phase Integral Methods and its Applications **December'24**
Prof. Ameeya Bhagwat, CEBS Mumbai *Ongoing Research Project*

- Explored the implications of differential equations, local analysis, and asymptotic approximations in modeling real-world systems such as plasma dynamics.
- Gained expertise in solving basic ordinary differential equations, advancing to asymptotic analysis techniques and the WKB approximation.

National Initiative for Undergraduate Sciences **June'24**
Tata Institute of Fundamental Research (TIFR), Mumbai *Research Exposure Camp*

- Selected among **72** students nationwide for a fortnight-long research exposure camp at the Homi Bhabha Centre for Science Education (HBCSE), attending seminars led by distinguished professors from institutions across the country.
- Gained a strong foundation in **theoretical**, **experimental**, and **computational** physics during the camp.
- Chosen for a year-long project on **"Phase Integral Method and its Applications"** under the guidance of **Professor Ameeya Bhagwat** (CBS Mumbai) due to outstanding performance in theoretical selection tests.

Scholastic Achievements

- Currently holding a **Department Rank 2** among over **60** students of the Engineering Physics'27
- Received the **Institute Award** given to **Top 50** among **1400+** students for stellar academic performance
- Achieved a Semester Performance Index **(SPI)** of **10/10** in Autumn semester of 1st year of undergraduate
- Secured an **All India Rank** of **487** in IIT-JEE Advanced examination among over **0.2 million** candidates
- Among **Top 48** nationwide in Indian National Mathematics Olympiad **(INMO)** and attended **IMOTC**
- Among **Top 35** nationwide in the Indian National Chemistry Olympiad **(INChO)** and attended **OCSC**
- Attained **Advanced Performer** (AP) grade in Entrepreneurship course awarded to top **2%** of the batch
- Secured an **All India Rank** of **436** in JEE Main examination among over **1.2+ million** candidates
- Among Top 1% students nationwide in National Standard Examination in Physics and selected for **INPhO**
- State Topper in National Standard Examination in Astronomy and selected for the **INAO**
- Recipient of the **KVPY** fellowship with an All India Rank **154**, offered admission to **IISc Bangalore**
- Awarded the **NTSE Scholarship** by NCERT, selected among **2000** students nationwide

Key Projects

Pinpricks on the Vortex - Star Cluster Analysis in Nearby Galaxies **May '24**
Krittika, The Astronomy Club, IIT Bombay *Summer Project*

- Analyzed the types and ages of the **star clusters** in the spiral galaxies using images from the **Hubble Telescope**
- Developed python scripts to perform **aperture photometry** and extract **cluster fluxes** in different wavelengths
- Correlated the flux differences to the physical properties of the star clusters such as **age and ellipticity**

Summer of Science - Elementary Particle Physics **May '24**
Maths and Physics Club, IIT Bombay *Summer Project*

- Explored **symmetry** properties, **Feynman diagrams**, and the basics of quantum field theory and their applications
- Conducted an in-depth study on the **Standard Model** and the associated **fundamental particle interactions**
- Explored foundational concepts in classical and quantum mechanics to enhance understanding of particle physics

Junior Design Engineer, Propulsion Subsystem, IIT Bombay Rocket Team **February '24**

IN-SPACe recognized technical team developing high-power rockets *Ongoing Technical Work*

- Designed **10+ solid-propellant rocket motors** in SolidWorks, optimizing propellants and motor performance
- Fabricated propellants with various compositions and conducted **Crawford Bomb Tests** to assess their properties
- Performed static fire tests on multiple manufactured rocket motors, achieving a total impulse exceeding 1000 $N \cdot s$
- Performed literature review on **hybrid motors**, ideating and researching on suitable fuels and methods

Turbulence: From Flowing Water to Molecular Clouds **September'24**

Prof. Punit Parmananda, IIT Bombay *Course Project*

- Investigated **non-linearity** in the **Navier-Stokes** equation, examining **turbulence** and chaotic behaviour of fluids
- Analyzed **vortex formation** in high Reynolds number flows, linking these dynamics to real-world phenomena
- Explored the impact of turbulence on various natural phenomena ranging from **atmospheric patterns to stars**

NetWeaver: Creating a Neural Network from Scratch **Dec '23**

Analytics Club, IIT Bombay *Winter Project*

- Developed **neural networks** from scratch to classify handwritten digits **(0-9)** with **94.96%** accuracy using MNIST
- Reviewed probability, statistics, and linear regression models to deepen understanding of neural network functions
- Applied **ReLU** activation and used **backpropagation** in a three-layer neural network for training and prediction

Other Projects

Introduction to Astronomy and Astrophysics **July '24**

Krittika, The Astronomy Club, IIT Bombay *Learner's Space*

- Gained an understanding of key astrophysics topics to enhance knowledge of stellar and galactic phenomena
- Started with fundamental concepts, including **coordinate systems and time**, celestial mechanics, and **telescopes**
- Advanced to specialized topics such as solar physics, **multi-messenger astronomy**, and stellar evolution

Universal Testing Machine **Nov'23**

Prof. Joseph John, Prof. Krishna Jonnalagadda, IIT Bombay *Course Project*

- Successfully designed and built a prototype of a **Universal Testing Machine** (UTM) for testing mechanical strength
- Utilized **AutoCAD** to ideate and create designs for the machine structure, gearbox, and specimen holder
- Developed an electric circuit using **Arduino** electronics for efficient data collection and automated gear operation

XLR8, Bot Making Competition **Aug '23**

Electronics and Robotics Club, IIT Bombay *Self Project*

- Developed a **hand-controlled robot** with a differential mechanism, enabling navigation over complex obstacles
- Utilized **ESP-32** microcontroller and transmitter, along with **L298 motor driver** and IMU, to achieve precise control
- Designed a robust and light robot chassis using AutoCAD, fabricating the chassis from acrylic sheets

Technical Skills

Programming Languages	C++, C, Python, MATLAB, Fortran, HTML, Markdown
Libraries	Astropy, NumPy, Scipy, Matplotlib, Pandas, Qiskit, QuTiP
ML and Data Science	Pytorch, Scikit-Learn, TensorFlow, Keras
Softwares	LaTeX, Git, LTSpice, Arduino, AutoCAD, OpenMotor, Ansys, ImageJ

Key Courses Undertaken

- Classical Mechanics
- Quantum Mechanics*
- Electromagnetism*
- General Relativity*
- Nonlinear Dynamics
- Thermal Physics
- AI and Data Science
- Abstract Algebra
- Calculus
- Differential Equations
- Oscillations and Waves
- Complex Analysis
- Statistical Mechanics*
- Analog Electronics
- Digital Electronics*
- Computer Programming

*To be completed by April '25

Volunteer Work

Volunteer and Core Team Member **Apr '24**
Maths and Physics Club, IIT Bombay *Ongoing Responsibility*

- Collaborating in a **15-member** team to organize and manage logistics for club events, including field trips, guest lectures, and group discussions, while playing a pivotal role in **creating problem statements and quiz questions**
- Created content and managed social media accounts of Facebook **(9K+ followers)** and Instagram **(2K+ followers)**

Extracurricular activities

- **Core team member** of quizzing contingent participating in Inter IIT Culturals 2024
- **Winner** of Astromania 2024, the Astronomy Quizzing Competition organised by Krittika at IITB
- Achieved 2nd **runner-up** in the **Convexing Complexity Challenge** organized by Enpower IIT Bombay
- Participated in **CricMania**, institute IPL-based cricket league organized by the Cricket Club at IITB
- Secured **first runner-up** at **Inquizitive: The Gen Quiz** during Avenues, organized by **SJMSOM, IITB**